CPEN 416

Lecture 09: Grover's algorithm

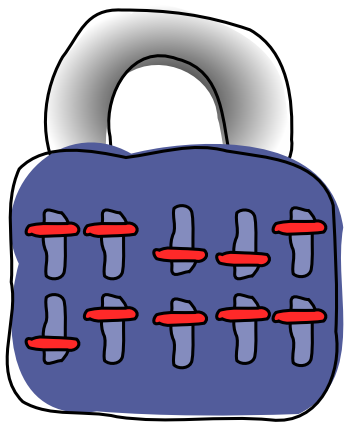
Tuesday 13 October 2026

Announcements

- Midterm available for pick up at my office
- Practice assignment 2 available
- Project details available later this week

Last time

We modeled breaking a (binary) lock as a function:



Trying combination $\mathbf{x}$ modeled as querying an *oracle* that evaluates $f(\mathbf{x})$. The number of queries is an algorithm's *query complexity*.

Image credit: Codebook node A.1

Last time

We expressed oracle queries as quantum circuits in two ways.

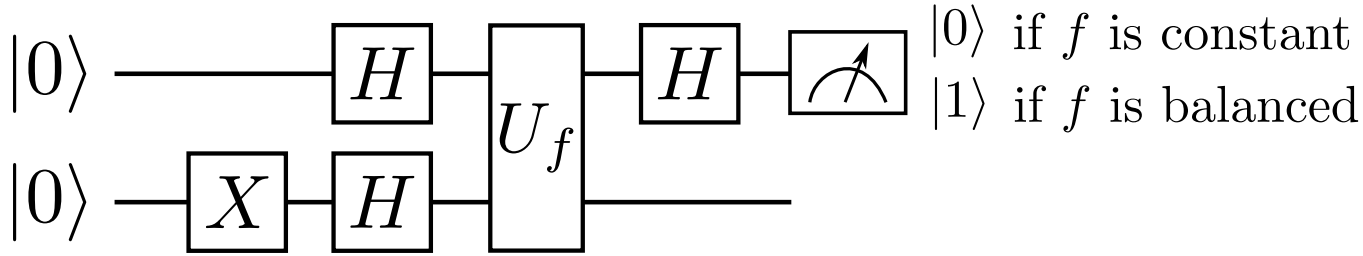
Last time

We saw how phase kickback can be used to obtain information about multiple evaluations of $f(x)$ with a single query.

Last time

We applied Deutsch's quantum algorithm to determine if a function is *constant* or *balanced* using one oracle query

Name	Action	Name	Action
f_1	$f_1(0) = 0$ $f_1(1) = 0$	f_2	$f_2(0) = 1$ $f_2(1) = 1$
f_3	$f_3(0) = 0$ $f_3(1) = 1$	f_4	$f_4(0) = 1$ $f_4(1) = 0$



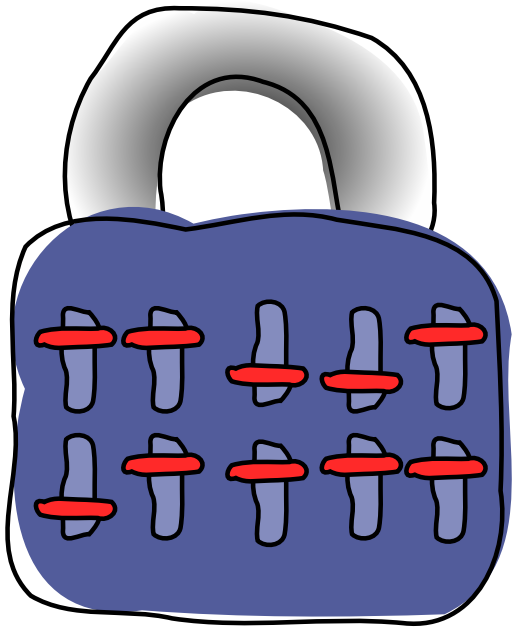
This is a quantum speedup; classical case requires 2 queries.

Learning outcomes

- Describe the strategy of amplitude amplification
- Derive the query complexity of Grover's search algorithm
- Implement Grover's search algorithm in PennyLane

Grover's quantum search algorithm

Let's break that lock!



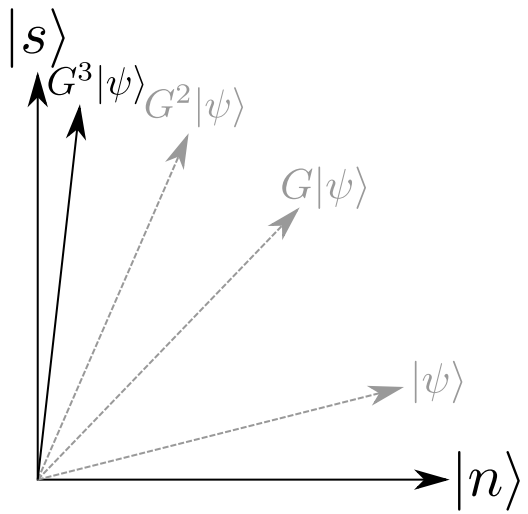
Classical: in the worst case, oracle queries

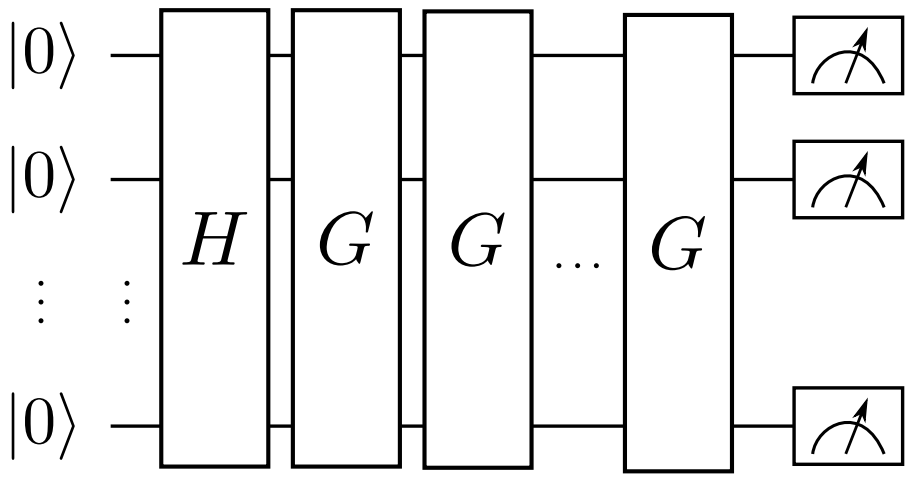
Quantum: queries with Grover's algorithm

Image credit: Codebook node A.1

Grover's quantum search algorithm

Grover's search algorithm starts with a uniform superposition, then *amplifies* the amplitude of the state corresponding to the solution.



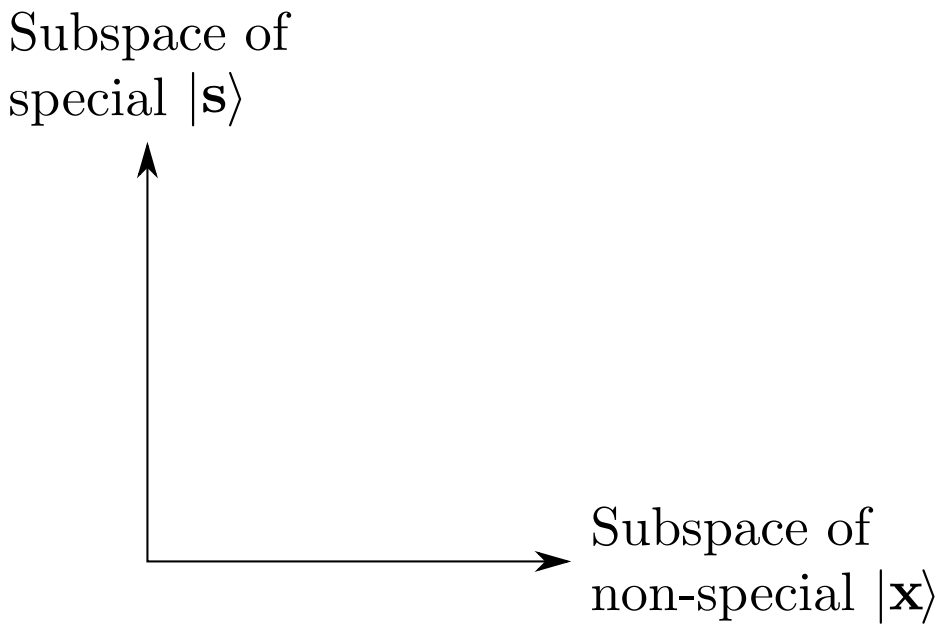


Grover's quantum search algorithm

In other words, we want to go from the uniform superposition

to something like this:

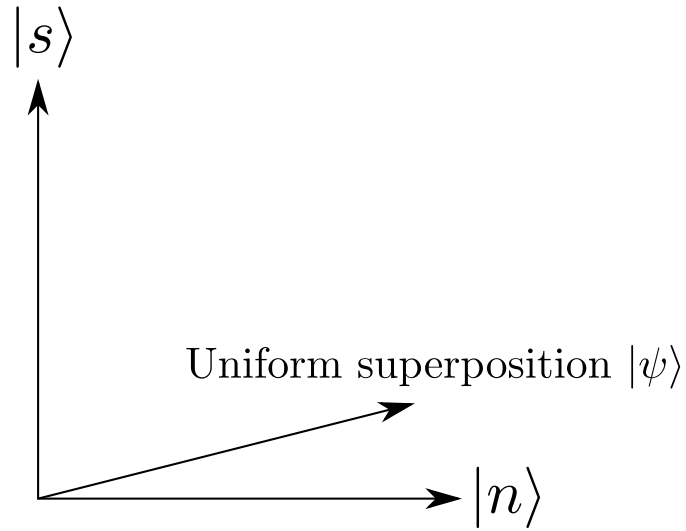
Amplitude amplification: geometric visualization



Partition the computational basis states into two subspaces:

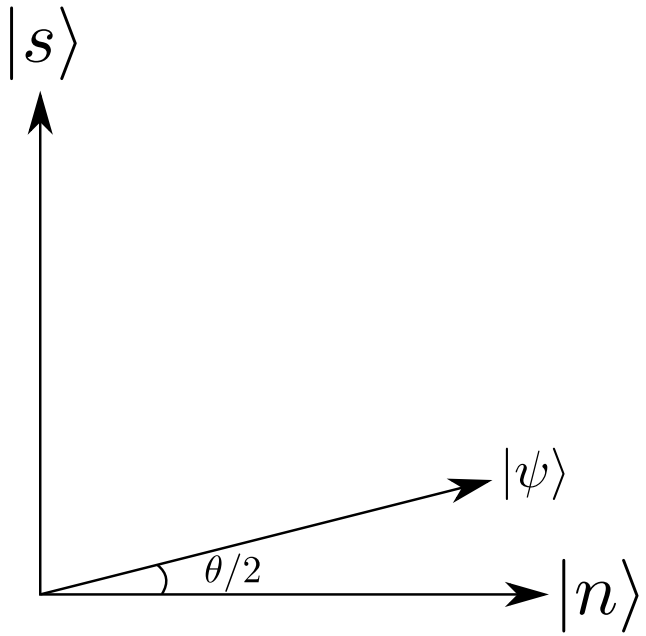
1. The special state $|\mathbf{s}\rangle$
2. All the other states

Amplitude amplification: geometric visualization



We can write the uniform superposition in terms of $|s\rangle$ and $|n\rangle$:

Amplitude amplification: geometric visualization

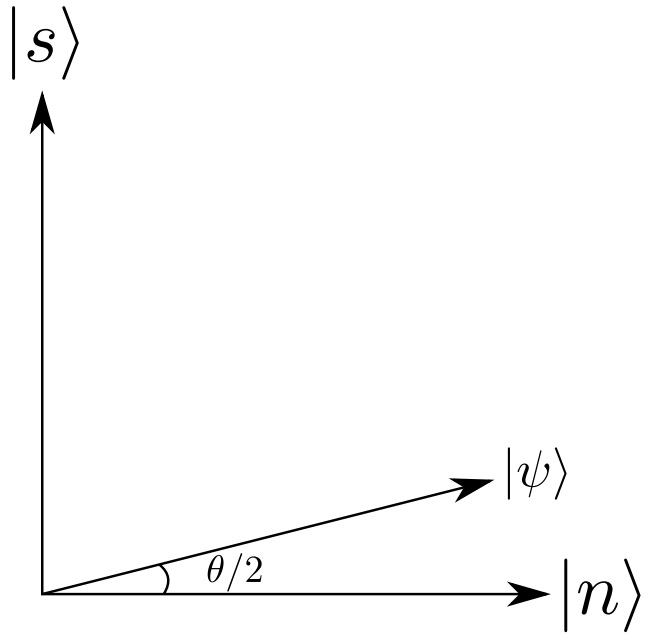


Instead of working with these complicated coefficients

$$|\psi\rangle = \frac{1}{\sqrt{2^n}}|s\rangle + \frac{\sqrt{2^n - 1}}{\sqrt{2^n}}|n\rangle,$$

rexpress in terms of an angle θ:

Amplitude amplification: geometric visualization

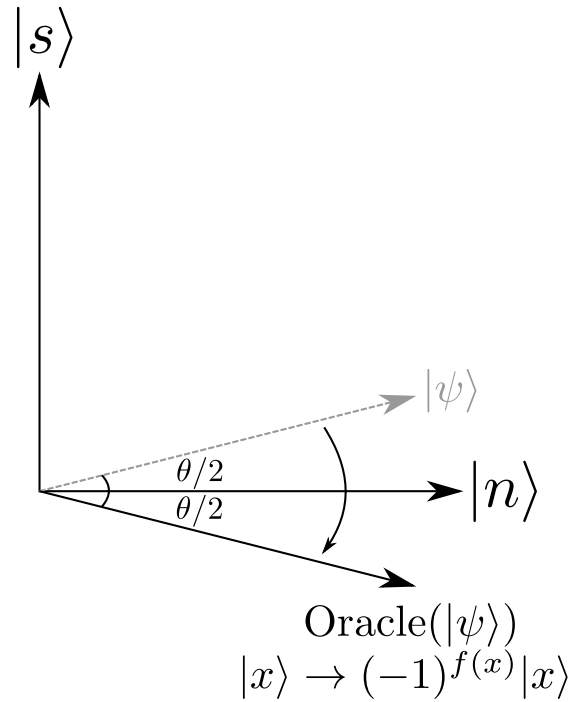


Apply operations to increase the amplitude of $|s\rangle$ while decreasing that of $|n\rangle$.

Two steps:

1. Apply the oracle O to 'pick out' the solution
2. Apply a 'diffusion operator' D to adjust the amplitudes.

Amplitude amplification: geometric visualization

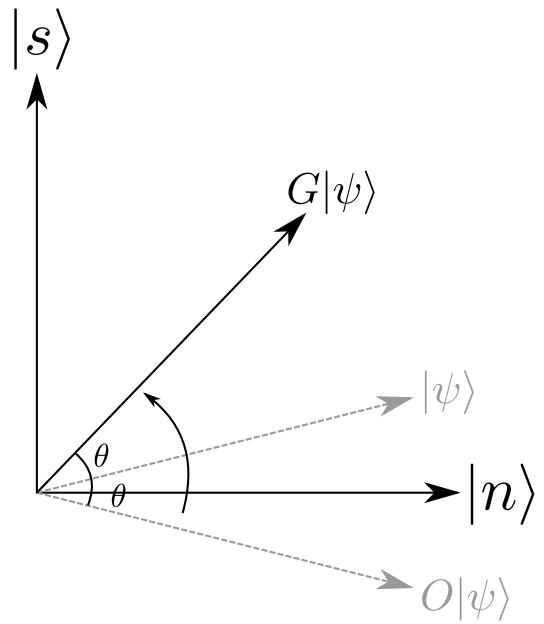


The oracle *flips* the amplitude of the special basis states in $|\psi\rangle$.

We can visualize this as a *reflection about the subspace* of non-special elements.

Exercise: how does this affect the uniform superposition state?

Amplitude amplification: geometric visualization

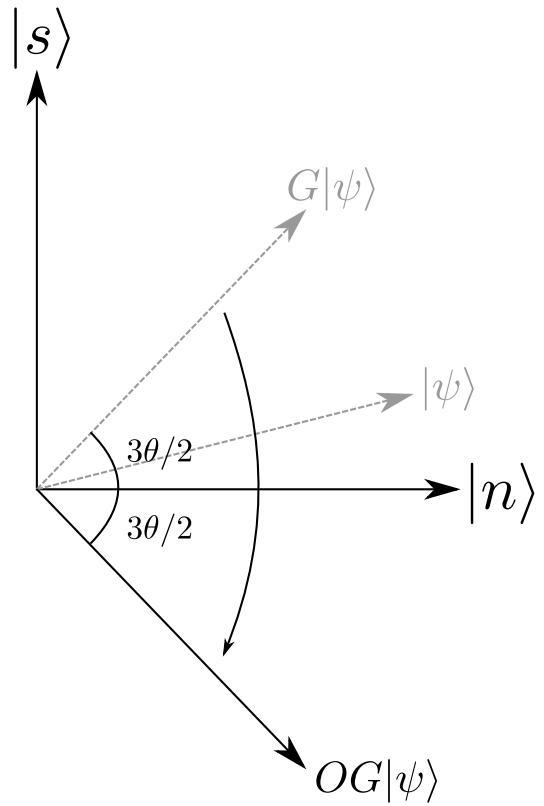


The diffusion operator is a bit less intuitive - it performs a *reflection about the uniform superposition* state.

Combined action $G = DO$ sends

$$G\left(\sin\left(\frac{\theta}{2}\right)|s\rangle + \cos\left(\frac{\theta}{2}\right)|n\rangle\right) =$$

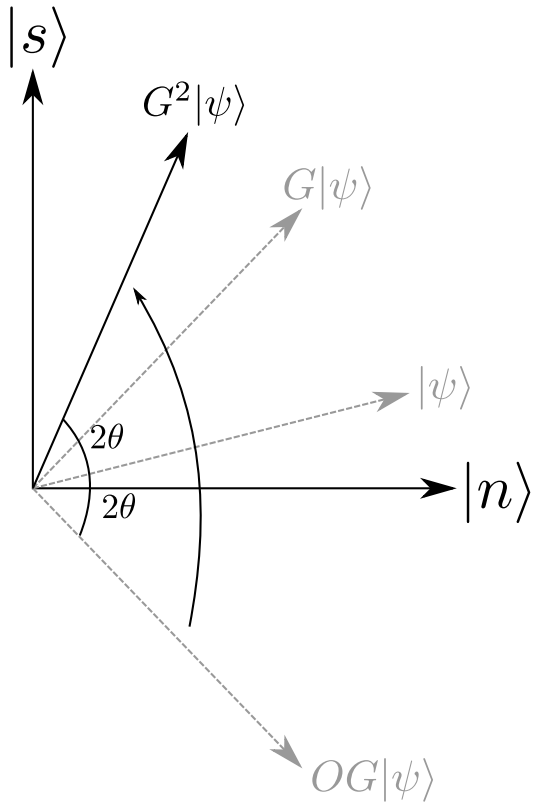
Grover's algorithm: geometric visualization



Grover's algorithm works by repeated application of $G = DO$ (called the Grover iteration).

Apply the oracle and reflect about the non-special elements.

Grover's algorithm: geometric visualization

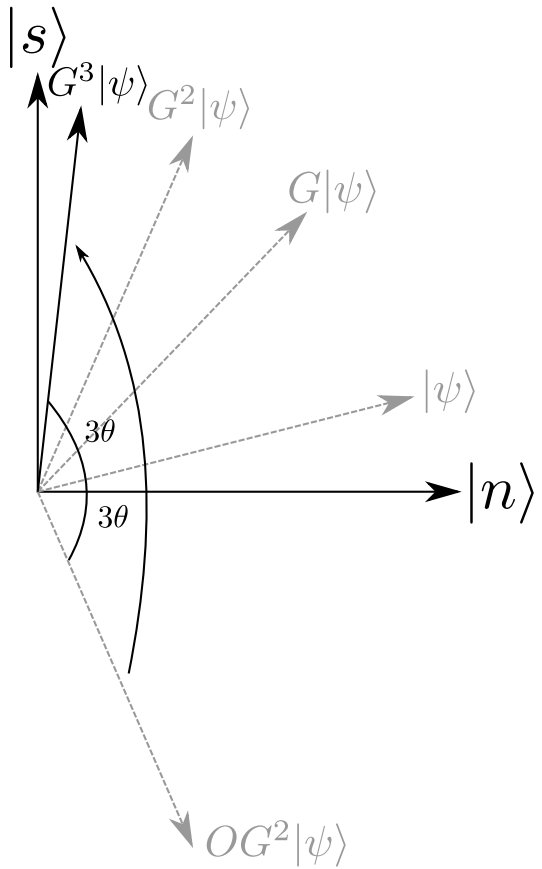


Apply the diffusion operator and reflect about the uniform superposition to boost the amplitude of the special state.

Grover's algorithm: geometric visualization

After k Grover iterations we will have the state

$$G^k|\psi\rangle =$$

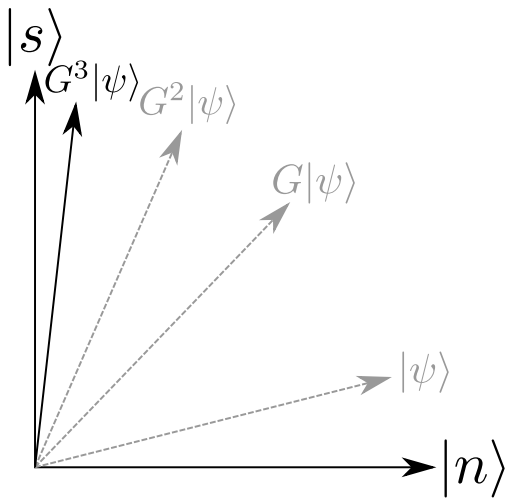


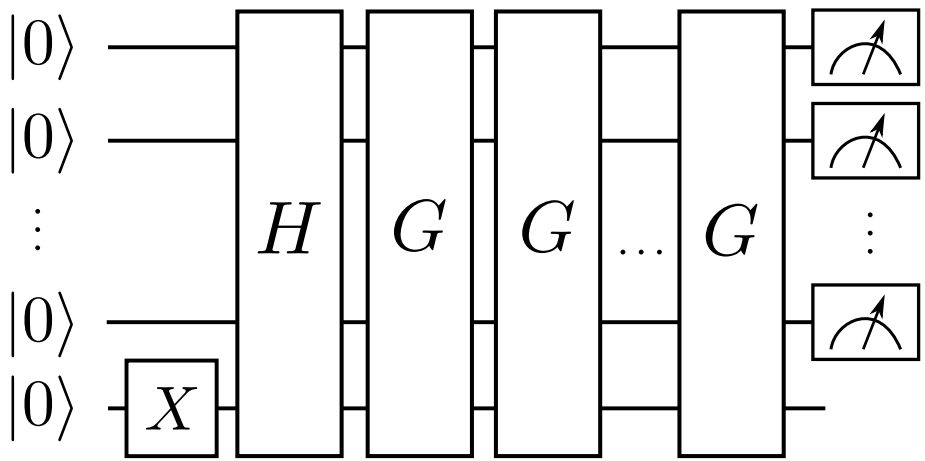
Exercise: what k *maximizes* the probability of measuring $|s\rangle$?

Implementing Grover search

We will use the "type 1" oracle:

$$O|\mathbf{x}\rangle|y\rangle = |\mathbf{x}\rangle|y \oplus f(\mathbf{x})\rangle$$

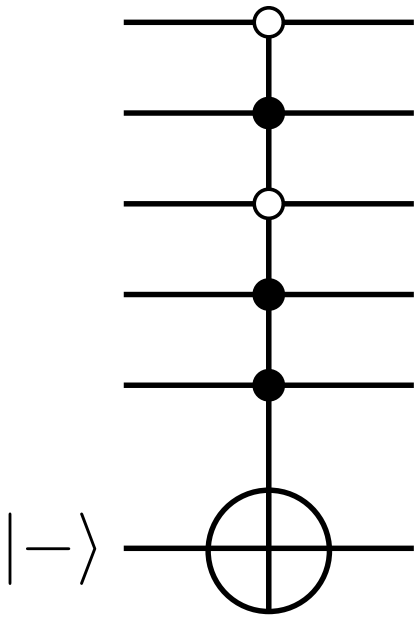




What do circuits for the oracle and diffusion look like?

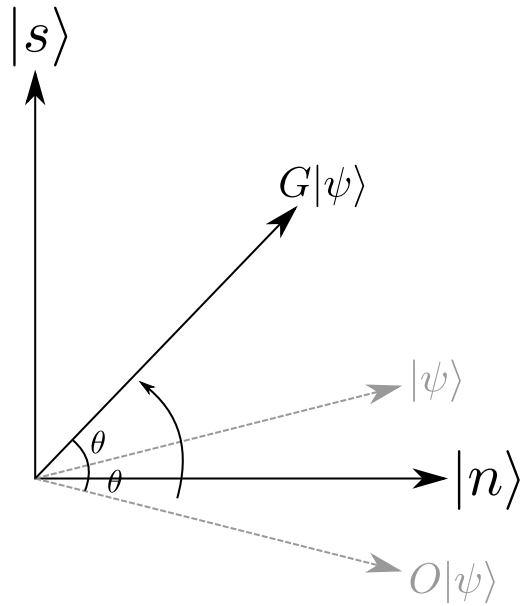
The oracle circuit

Exercise: show that a multicontrolled X gate, controlled on **s**, can be used as an oracle:



The diffusion circuit

The diffusion operator performs a reflection about the uniform superposition state.



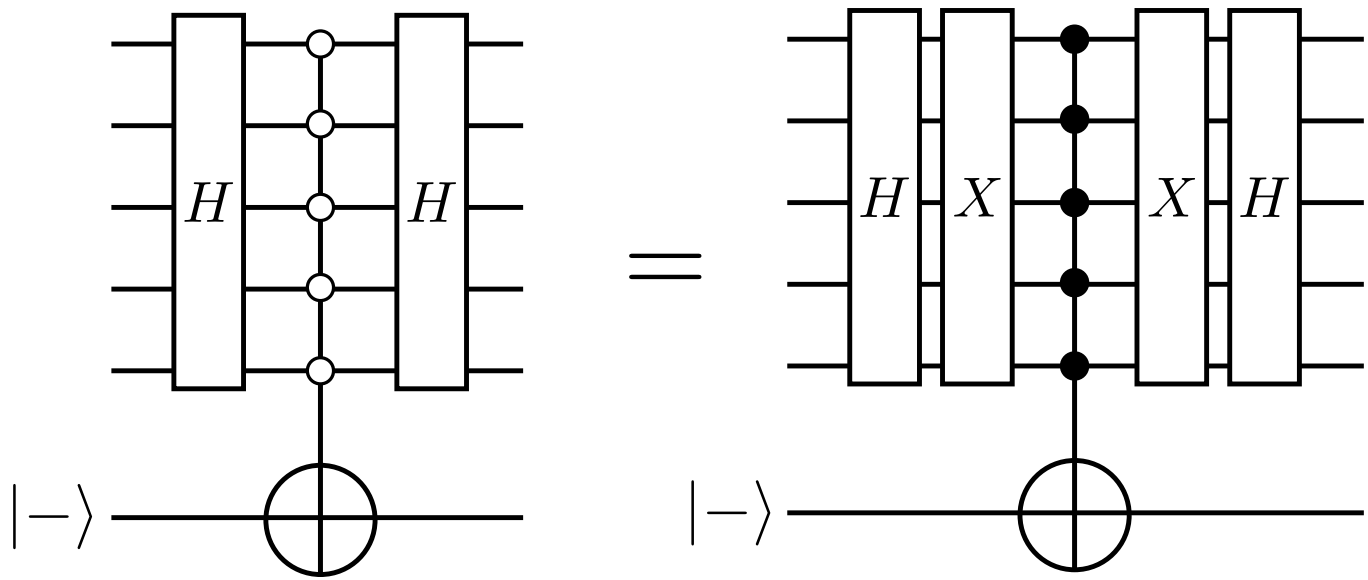
The diffusion circuit

Exercise: Show that the unitary matrix given by

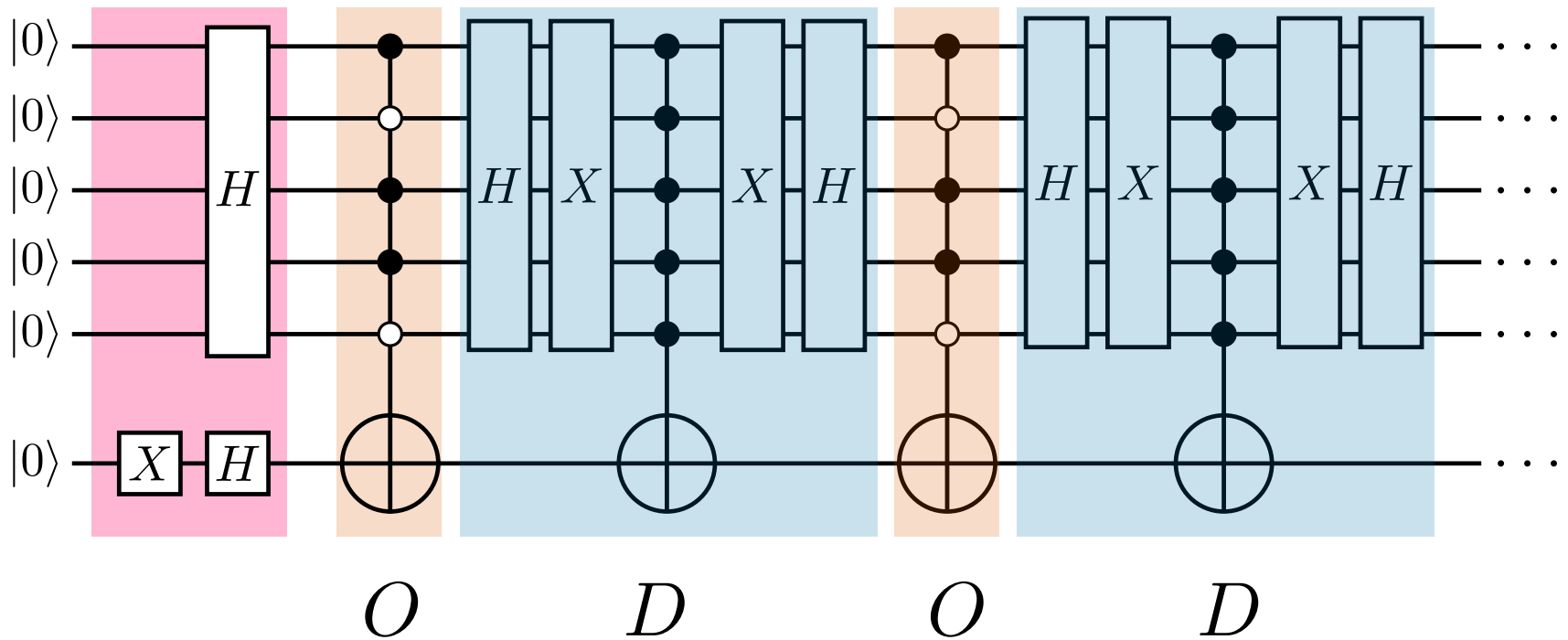
$$D = 2|\psi\rangle\langle\psi| - I$$

is equivalent to the diffusion operator.

The diffusion circuit

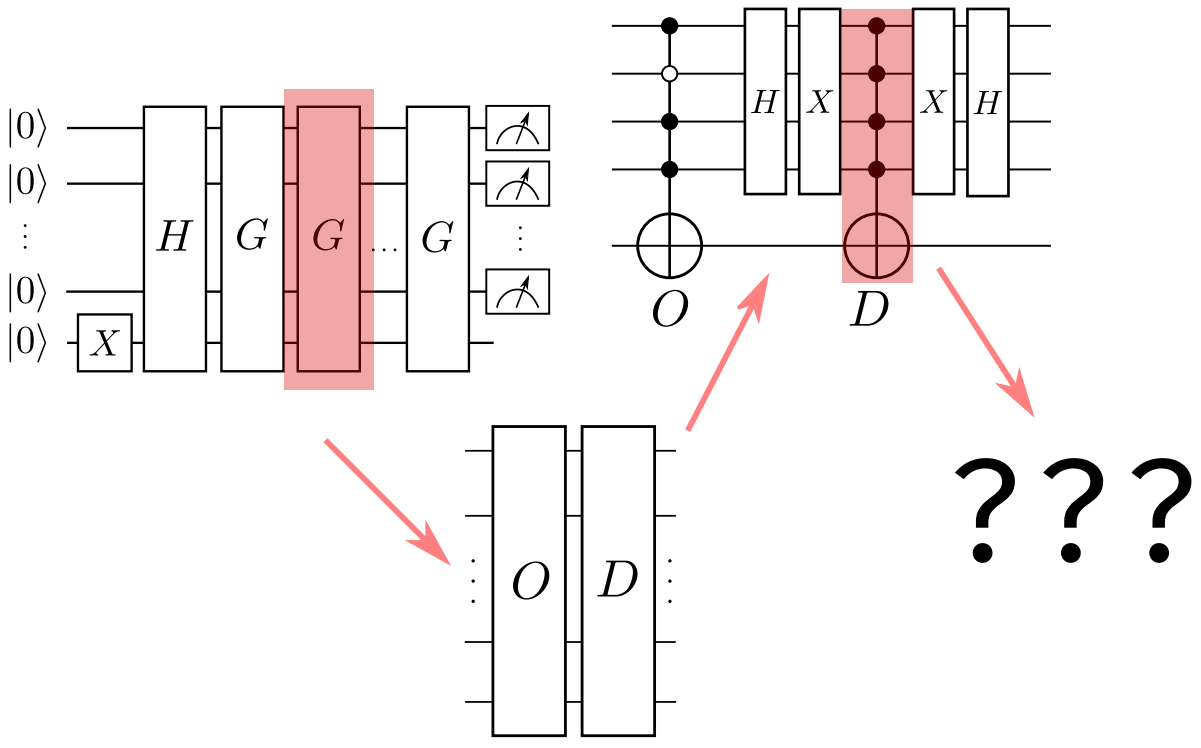


The full Grover circuit



The full Grover circuit

Clearly, each of the $O(\sqrt{2^n})$ queries requires some number of gates... how much does Grover *really* cost?



Next class: look inside the black box!

Next time

Content:

- Introduction to quantum compilation and resource estimation

Action items:

1. Start picking project group
2. Practice assignment 2

Recommended reading:

- For this class: Codebook module GA; Nielsen & Chuang 6.1
- For next class: Nielsen & Chuang 4.5